
📚 Lab 1: Introduction to SCADA/ICS (Easy Notes)

🎯 Objectives

After completing this lab, you will understand:

- What is **SCADA**?
- What is **ICS**?
- Where are they used?
- Why is their security important?

🏭 What is ICS?

ICS stands for **Industrial Control Systems**.

Simple Definition

ICS is a collection of systems used to **control machines and industrial processes**.

Real-Life Example

Imagine a **factory** that makes bottled water.

Many machines work together:

- Filling bottles
- Putting caps on bottles
- Labeling bottles
- Packing bottles

Someone or something has to control all these machines.

➡️ That complete control system is called **ICS**.

Think of ICS as the **brain of a factory**.

🖥️ What is SCADA?

SCADA stands for:

Supervisory Control And Data Acquisition

Simple Definition

SCADA is software that:

- Watches industrial machines
- Collects live (real-time) data
- Sends commands to machines
- Shows everything on a computer screen

Real-Life Example

Imagine an electricity company.

The operator sits in a control room.

On the computer, they can see:

- Voltage
- Temperature
- Power usage
- Alarms
- Equipment status

If something goes wrong, they can remotely turn equipment on or off.

That is **SCADA**.

🧠 **Easy Difference**

ICS	**SCADA**
Complete industrial control system	One type of industrial control system
Controls machines	Monitors and controls machines remotely
Includes PLC, DCS and SCADA	Mainly collects data and supervises

Remember:

Every SCADA is part of ICS, but not every ICS is SCADA.

Types of ICS

There are **three main types**.

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1. PLC

PLC = **Programmable Logic Controller**

A small industrial computer.

It controls one machine.

Example:

A traffic light.

It automatically changes:

Red → Yellow → Green

A PLC controls this.

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2. DCS

DCS = **Distributed Control System**

Used inside a large factory.

Many controllers work together.

Example:

An oil refinery.

Different sections:

- Heating
- Cooling
- Mixing

Each section has its own controller.

Together they form a DCS.

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3. SCADA

SCADA is used when machines are spread over large distances.

Example:

A city water supply.

The water pumps are many kilometers apart.

SCADA monitors and controls them from one control room.

Where is SCADA/ICS Used?

⚡ **Energy Sector**

Power plants

Example:

- Electricity generation
- Power distribution

💧 **Water Treatment**

Water plants use SCADA to:

- Check water quality
- Turn pumps ON/OFF
- Monitor water levels

🏭 **Manufacturing**

Factories use ICS for automation.

Example:

Car manufacturing

Robotic arms automatically:

- Weld
- Paint
- Assemble

🛢️ **Oil & Gas**

Used to control:

- Pipelines
- Pumps
- Refineries

Why is SCADA Security Important?

Imagine someone hacks a power plant.

Possible problems:

❌ Electricity shuts down

❌ Water supply stops

❌ Factory production stops

❌ Explosion

❌ Environmental damage

❌ Human injuries

That's why SCADA security is extremely important.

Example: Stuxnet

One of the most famous cyber attacks.

What happened?

Hackers created malware called **Stuxnet**.

It attacked Iran's nuclear plant.

Instead of destroying computers,

it secretly changed the speed of industrial machines.

Many machines were damaged.

This showed the whole world that SCADA and ICS can be hacked.

How to Protect SCADA/ICS?

1. Network Segmentation

Separate the industrial network from the office network.

Example:

Office PCs

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Firewall

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Industrial machines

This way hackers cannot easily reach the industrial systems.

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2. Regular Updates

Always install:

- Security patches
- Software updates

to fix vulnerabilities.

Case Study

Water Treatment Plant

SCADA monitors:

- Water level
- Water pressure
- Chemical balance
- Pumps

If the water level becomes low,

SCADA automatically starts the pump.

Easy Summary

Term	Easy Meaning
ICS	Complete industrial control system
SCADA	Software that monitors and controls industrial processes
PLC	Small industrial computer
DCS	Controls one complete factory
SCADA	Controls machines over long distances

⭐ **Important Interview Question**

Q: Difference between SCADA and ICS?

Answer:

ICS is a broad term for industrial control systems. It includes PLCs, DCS, and SCADA. SCADA is a type of ICS used for monitoring, collecting real-time data, and controlling industrial processes remotely.
